

SecOps AI Monitoring Stack Architecture

SecOps AI Monitoring Stack Architecture, designed to enhance security operations by integrating AI capabilities, specifically Large Language Models (LLM), into the monitoring stack. This architecture aims to streamline incident response, threat detection, and system health monitoring.

1. Data Flow

The data flow within the SecOps AI Monitoring Stack is structured as follows:

flowchart TD

%% Application Stack Components

subgraph AppStack [Application Stack]

WP[WordPress Container] --> PT[Promtail]

NG[Nginx Proxy Container] --> PT

DB[MySQL Container] --> PT

end

%% Log Collection & Processing

PT --> LK[(Grafana Loki)]

ME[MySQL Exporter] --> PR[Prometheus]

NE[Node Exporter] --> PR

NXE[Nginx Exporter] --> PR

%% AI Analysis Pipeline

subgraph AI_Analysis [AI Analysis Pipeline]

AA[AI Adapter] --> 1. LK [60s]

LK --> 2. AA

AA --> 3. OL[Ollama: Llama-3 GPU]

OL --> 4. JSON

AA --> 5. AM[Alertmanager]

end

%% Alerting & Reporting

PR --> AM

%% Summary & Output

3. **LLM (Large Language Model)**

LLM (Llama-3) ■ ■■■ ■■■■■■■■■■■■ ■■■■■■■ ■■■ ■■■■■■■■■■ ■■■■■■■■:

■. ■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■ (Streaming Log Scanning)

AI-[REDACTED] ([REDACTED]) [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED]. [REDACTED] [REDACTED] [REDACTED] [REDACTED]: * [REDACTED] SQL-[REDACTED]
(SQLi) * [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] (RCE) * [REDACTED] [REDACTED]
brute-force ([REDACTED] [REDACTED]) * [REDACTED] [REDACTED] [REDACTED] (XSS)

[illegible][illegible]

■. ■■■■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ (On-the-fly Alert Enrichment)

```
##### (#####, ##### MySQL
#####), #####
##### MySQL database is DOWN.
```

```

##### LLM: 1. #####. 2. #####
##### Loki ##### 10-15 #####
#####. 3. ##### Llama-3 #####
#####. 4. ##### (#####
##### Out of memory: Killed process ##### Access denied for user) #####
#####: * ##### (##### Out of Memory
Crash). * ##### (Summary) #####, #####, #####
#####. * ##### (Mitigation Steps) (#####
#####'##### compose-#####
#####).
```

[illegible]